LAB REPROT

LAB: CLASSIFICATION DATA - 3

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Bài làm

Exercise 4: Using Java programming language to construct a decision tree with the J-48 algorithm (or the other classification model) for the above exercises

1. Load dữ liệu "weather.nominal.arff"

PreprocessData.java | ClassificationData.java | Demo.java

```java
package pj_DataMiningLabs;

import weka.core.Instances;
import weka.core.converters.ConverterUtils.DataSource;

public class PreprocessData {
    DataSource source;
    Instances dataset;

    public void loadData(String fileName) throws Exception {
        source = new DataSource(fileName);
        dataset = source.getDataSet();
    }

    public void outputData() {
        System.out.println(dataset.toSummaryString());
        System.out.println(dataset.toString());
    }
}
```

Kết quả demo

PreprocessData.java | ClassificationData.java | Demo.java

```java
package pj_DataMiningLabs;

public class Demo {

    public static void main(String[] args) throws Exception {
        PreprocessData preprocess = new PreprocessData();

        preprocess.loadData("D:\\Weka-3-9-6\\data\\weather.nominal.arff");
        preprocess.outputData();
```

Problems | Javadoc | Declaration | Console | Install Java 26 Support

```
<terminated> Demo [Java Application] C:\Users\NGUYEN ANH HUY\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-
Relation Name:  weather.symbolic
Num Instances:  14
Num Attributes: 5

     Name                     Type  Nom  Int Real     Missing      Unique  Dist
1 outlook                    Nom 100%   0%   0%     0 /  0%     0 /  0%     3
2 temperature                Nom 100%   0%   0%     0 /  0%     0 /  0%     3
3 humidity                   Nom 100%   0%   0%     0 /  0%     0 /  0%     2
4 windy                      Nom 100%   0%   0%     0 /  0%     0 /  0%     2
5 play                       Nom 100%   0%   0%     0 /  0%     0 /  0%     2

@relation weather.symbolic

@attribute outlook {sunny,overcast,rainy}
@attribute temperature {hot,mild,cool}
@attribute humidity {high,normal}
@attribute windy {TRUE,FALSE}
@attribute play {yes,no}

@data
sunny,hot,high,FALSE,no
sunny,hot,high,TRUE,no
overcast,hot,high,FALSE,yes
rainy,mild,high,FALSE,yes
rainy,cool,normal,FALSE,yes
rainy,cool,normal,TRUE,no
overcast,cool,normal,TRUE,yes
sunny,mild,high,FALSE,no
sunny,cool,normal,FALSE,yes
```

2. Thực hiện phân lớp với thuật toán J48: “Cross-validation” (10 folds) test mode

- Code cài đặt và kết quả in model

PreprocessData.java | ClassificationData.java | Demo.java

```java
package pj_DataMiningLabs;

import weka.classifiers.bayes.NaiveBayes;
import weka.classifiers.rules.OneR;
import weka.classifiers.trees.J48;
import weka.core.Instances;

public class ClassificationData {
    J48 j48Model;
    OneR oneRModel;
    NaiveBayes nbModel;

    public void buildJ48Model(Instances datatrain, float cf) throws Exception {
        datatrain.setClassIndex(datatrain.numAttributes() - 1);
        j48Model = new J48();
        j48Model.setConfidenceFactor(cf);
        j48Model.buildClassifier(datatrain);
    }

    public String outputJ48Model() {
        return j48Model.toString();
    }
}
```

```java
package pj_DataMiningLabs;

import weka.core.Instances;

public class Demo {

    public static void main(String[] args) throws Exception {
        Instances datatrain;
        PreprocessData preprocess = new PreprocessData();

        datatrain = preprocess.loadData("D:\\Weka-3-9-6\\data\\weather.nominal.arff");

        // preprocess.outputData();
        ClassificationData classify = new ClassificationData();
        classify.buildJ48Model(datatrain, 0.35f);
        System.out.println(classify.outputJ48Model());

    }

}
```

```
<terminated> Demo [Java Application] C:\Users\NGUYEN ANH HUY\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin\
J48 pruned tree
------------------

outlook = sunny
|   humidity = high: no (3.0)
|   humidity = normal: yes (2.0)
outlook = overcast: yes (4.0)
outlook = rainy
|   windy = TRUE: no (2.0)
|   windy = FALSE: yes (3.0)

Number of Leaves  :     5

Size of the tree :      8
```

- Đánh giá mô hình

```java
public void evalJ48Model(Instances datatest) throws Exception {
    datatest.setClassIndex(datatest.numAttributes() - 1);

    evalModel = new Evaluation(datatest);
    evalModel.evaluateModel(j48Model, datatest);

    System.out.println("=== Evaluation Result ===");
    System.out.println(evalModel.toSummaryString());

    System.out.println("=== Confusion Matrix ===");
    System.out.println(evalModel.toMatrixString());
```

```
=== Evaluation Result ===

Correctly Classified Instances          14               100      %
Incorrectly Classified Instances         0                 0      %
Kappa statistic                          1
Mean absolute error                      0
Root mean squared error                  0
Relative absolute error                  0      %
Root relative squared error              0      %
Total Number of Instances               14

=== Confusion Matrix ===
=== Confusion Matrix ===

 a b   <-- classified as
 9 0 | a = yes
 0 5 | b = no
```

3. Thực hiện phân lớp với thuật toán OneR

- Code cài đặt

```java
    public void buildOneRModel(Instances datatrain) throws Exception {
        datatrain.setClassIndex(datatrain.numAttributes() - 1);

        oneRModel = new OneR();
        oneRModel.buildClassifier(datatrain);
    }

    public String outputOneRModel() {
        return oneRModel.toString();
    }


    public void evalOneRModel(Instances datatest) throws Exception {
        datatest.setClassIndex(datatest.numAttributes() - 1);

        evalModel = new Evaluation(datatest);

        evalModel.evaluateModel(oneRModel, datatest);

        System.out.println("=== OneR Evaluation ===");
        System.out.println(evalModel.toSummaryString());
        System.out.println(evalModel.toMatrixString());
    }

}
```

- In kết quả áp dụng model

```java
        // OneR
        classify.buildOneRModel(datatrain);

        // In model
        System.out.println(classify.outputOneRModel());

        // Evaluate
        classify.evalOneRModel(datatrain);
    }


}
```

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<terminated> Demo [Java Application] C:\Users\NGUYEN ANH HUY\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin

```
outlook:
        sunny    -> no
        overcast-> yes
        rainy    -> yes
(10/14 instances correct)

=== OneR Evaluation ===

Correctly Classified Instances          10               71.4286 %
Incorrectly Classified Instances         4               28.5714 %
Kappa statistic                          0.3778
Mean absolute error                      0.2857
Root mean squared error                  0.5345
Relative absolute error                 61.5385 %
Root relative squared error            111.4773 %
Total Number of Instances               14

=== Confusion Matrix ===

 a b   <-- classified as
 7 2 | a = yes
 2 3 | b = no
```

4. Thực hiện thuật toán NaiveBayes

- Code cài đặt

```java
public void buildNBModel(Instances datatrain) throws Exception {
    datatrain.setClassIndex(datatrain.numAttributes() - 1);

    nbModel = new NaiveBayes();
    nbModel.buildClassifier(datatrain);
}

public String outputNBModel() {
    return nbModel.toString();
}

public void evalNBModel(Instances datatest) throws Exception {
    datatest.setClassIndex(datatest.numAttributes() - 1);

    evalModel = new Evaluation(datatest);

    evalModel.evaluateModel(nbModel, datatest);

    System.out.println("=== Naive Bayes Evaluation ===");
    System.out.println(evalModel.toSummaryString());
    System.out.println(evalModel.toMatrixString());
}
```

- Kết quả áp dụng model

```java
        // naive Bayes
        classify.buildNBModel(datatrain);
        System.out.println(classify.outputNBModel());
        classify.evalNBModel(datatrain);
```

Problems | Javadoc | Declaration | Console | Install Java 26 Support

```
<terminated> Demo [Java Application] C:\Users\NGUYEN ANH HUY\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21

                Class
Attribute         yes     no
                (0.63) (0.38)
============================
outlook
  sunny           3.0    4.0
  overcast        5.0    1.0
  rainy           4.0    3.0
  [total]        12.0    8.0

temperature
  hot             3.0    3.0
  mild            5.0    3.0
  cool            4.0    2.0
  [total]        12.0    8.0

humidity
  high            4.0    5.0
  normal          7.0    2.0
  [total]        11.0    7.0

windy
  TRUE            4.0    4.0
  FALSE           7.0    3.0
  [total]        11.0    7.0
```

```
=== Naive Bayes Evaluation ===

Correctly Classified Instances          13               92.8571 %
Incorrectly Classified Instances         1                7.1429 %
Kappa statistic                          0.8372
Mean absolute error                      0.2917
Root mean squared error                  0.3392
Relative absolute error                 62.8233 %
Root relative squared error             70.7422 %
Total Number of Instances               14

=== Confusion Matrix ===

 a b   <-- classified as
 9 0 | a = yes
 1 4 | b = no
```

5. Thực hiện Train-Test Split với 70%

- Code cài đặt

```java
    public Instances createTrainTestSet(Instances data, boolean isTrain, double per) throws Exception {
        RemovePercentage rm = new RemovePercentage();

        rm.setPercentage(per);
        rm.setInvertSelection(isTrain);
        rm.setInputFormat(data);

        return Filter.useFilter(data, rm);
    }

    public void outputDataset() {
        System.out.println(dataset.toSummaryString());
    }
}
```

- Kết quả

```java
    public static void main(String[] args) throws Exception {
        Instances datatrain, dataset, datatest;
        PreprocessData preprocess = new PreprocessData();

        dataset = preprocess.loadData("D:\\Weka-3-9-6\\data\\weather.nominal.arff");

        datatrain = preprocess.createTrainTestSet(dataset, true, 70);
        ClassificationData classify = new ClassificationData();
        classify.buildJ48Model(datatrain, 0.35f);
        System.out.println(classify.outputJ48Model());

        datatest = preprocess.createTrainTestSet(dataset, false, 70);
        System.out.println("============= TẬP TEST =================");
        for (int i = 0; i < datatest.numInstances(); i++)
            System.out.println(datatest.instance(i).toString());

        System.out.println("============= KẾT QUẢ ĐÁNH GIÁ MÔ HÌNH ==========");
        classify.evalJ48Model(datatest);
```

Problems | Javadoc | Declaration | Console | Install Java 26 Support

```
<terminated> Demo [Java Application] C:\Users\NGUYEN ANH HUY\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin\javaw.exe (Mar 24, 202
J48 pruned tree
------------------

outlook = sunny: no (4.0/1.0)
outlook = overcast: yes (2.0)
outlook = rainy: yes (4.0/1.0)

Number of Leaves  :     3

Size of the tree :      4

============= TẬP TEST =================
sunny,mild,normal,TRUE,yes
overcast,mild,high,TRUE,yes
overcast,hot,normal,FALSE,yes
rainy,mild,high,TRUE,no
```

```
============= KET QUA DANH GIA MO HINH ==========
=== Evaluation Result ===

Correctly Classified Instances          2               50      %
Incorrectly Classified Instances        2               50      %
Kappa statistic                        -0.3333
Mean absolute error                     0.375
Root mean squared error                 0.5303
Relative absolute error                90      %
Root relative squared error           120.2676 %
Total Number of Instances               4

=== Confusion Matrix ===
=== Confusion Matrix ===

 a b   <-- classified as
 2 1 | a = yes
 1 0 | b = no
```

6. Thực hiện prediction

- Code cài đặt

```java
    public void predictLabel(Instances datapredict, String fileout) throws Exception {
        datapredict.setClassIndex(datapredict.numAttributes() - 1);
        for (Instance inst: datapredict) {
            double label = j48Model.classifyInstance(inst);
            inst.setClassValue(label);
            System.out.println(inst.toString());
        }

        BufferedWriter bw = new BufferedWriter(new FileWriter(fileout));
        bw.write(datapredict.toString());
        bw.close();

    }
}
```

- Kết quả

```java
        classify.predictLabel(datatest, "E:\\DataMining\\pj_DataMiningLabs\\output.arff");

        System.out.println("Prediction đã được lưu trong output.arff");

        /*
        // preprocess.outputData();
```

Problems | Javadoc | Declaration | Console | Install Java 26 Support

<terminated> Demo [Java Application] C:\Users\NGUYEN ANH HUY\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v2026

```
sunny,mild,normal,TRUE,no
overcast,mild,high,TRUE,yes
overcast,hot,normal,FALSE,yes
rainy,mild,high,TRUE,yes
Prediction đã được lưu trong output.arff
```

note.txt | link_repo.txt | student.txt | buy_comput | buy_comp_e | outpu

File | Edit | View

```
@relation weather.symbolic-weka.filters.unsupervised.instance.RemovePercentage-P70.0

@attribute outlook {sunny,overcast,rainy}
@attribute temperature {hot,mild,cool}
@attribute humidity {high,normal}
@attribute windy {TRUE,FALSE}
@attribute play {yes,no}

@data
sunny,mild,normal,TRUE,no
overcast,mild,high,TRUE,yes
overcast,hot,normal,FALSE,yes
rainy,mild,high,TRUE,yes
```